Modern Academy
for Engineering and Technology in Maadi
Basic Science Department

MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

Academic Year	2025-2026
Semester	First
Exam Date	13/11/2025
Academic Level:	Sophomore

MODEL ANSWER FOR THE MID-TERM WRITTEN EXAM

Subject: (MTHn103) **Mathematics-3** Differential Equations and Transforms
Spec.: Electrical and Mechanical Engineering
Examiners: Prof. Ashraf T.EL-Sayed and Dr. Amira M.Salem
Time: 60 minutes

Number of Pages: 1 **(Both Sides)** Number of Questions: 2 Attempt all questions

Question 1: [LO's: LO 1, LO 2, LO 4] **(12 points)**

Find the general solution of the first ordinary differential equations:

a) $\left(4+e^{2x}\right)y\dfrac{dy}{dx}=e^{2x-y^2}$. **(3 points)**

$$\boxed{\frac{1}{2}}\int e^{y^2}\boxed{2}\,y\,dy=\boxed{\frac{1}{2}}\int\frac{\boxed{2}\,e^{2x}}{\left(4+e^{2x}\right)}dx$$

$$\boxed{\frac{1}{2}e^{y^2}=\frac{1}{2}\ln\left|4+e^{2x}\right|+C}$$

b) $xy'-y=x^2e^{\left(\frac{y}{x}\right)}$. **(3 points)**

$$y'-\frac{y}{x}=xe^{\left(\frac{y}{x}\right)} \qquad \boxed{\frac{y}{x}=u\rightarrow y=xu\rightarrow y'=xu'+u}$$

$$x\frac{du}{dx}+\cancel{u}-\cancel{u}=xe^{u}\rightarrow\int e^{-u}du=\int dx$$

$$-e^{-u}=x+C$$

$$\boxed{-e^{-\left(\frac{y}{x}\right)}=x+C}$$

c) $\left(xy+y^2\right)dx+\left(x+2y-1\right)dy=0$. **(3 points)**

$$M_y=x+2y \qquad N_x=1\rightarrow M_y\neq N_x$$

$$M_y-N_x=x+2y-1\rightarrow\frac{M_y-N_x}{N}=1\rightarrow\mu=e^{\int dx}=e^{x}$$

$$\left(xye^{x}+y^2e^{x}\right)dx+\left(xe^{x}+2ye^{x}-e^{x}\right)dy=0\rightarrow\int\left(xye^{x}+y^2e^{x}\right)dx+\int\left(xe^{x}+2ye^{x}-e^{x}\right)dy=C$$

$$xye^{x}-ye^{x}+y^2e^{x}+\cancel{xe^{x}y}+\cancel{y^2e^{x}}-\cancel{e^{x}y}=C\rightarrow\boxed{xye^{x}-ye^{x}+y^2e^{x}=C}$$

d) $xy'-y=\left(x^2e^{x}+x^5\right)$. **(3 points)**

$$y'-\frac{1}{x}y=\left(xe^{x}+x^4\right)$$

$$\mu=e^{\int-\frac{1}{x}dx}=e^{-\ln x}=e^{\ln x^{-1}}=x^{-1}$$

$$y=x\left[\int\left(e^{x}+x^3\right)dx+C\right]$$

$$\boxed{y=x\left[\left(e^{x}+\frac{x^4}{4}\right)+C\right]}$$

Question 2: [LO's: LO 1, LO 2, LO 4] **(8 points)**

a) Solve the differential equation:

$$\frac{d^6y}{dx^6}+5\frac{d^5y}{dx^5}+9\frac{d^4y}{dx^4}+6\frac{d^3y}{dx^3}=0$$

(4 points)

$$m^6+5m^5+9m^4+6m^3=0$$

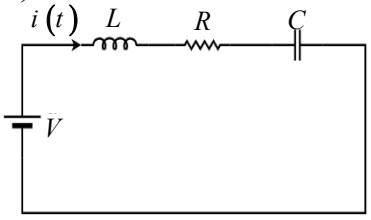
$$m^3\left(m^3+5m^2+9m+6\right)=0$$

$$m_1=m_2=m_3=0, m_4=-2, m_{5,6}=-\frac{3}{2}\pm i\,\frac{\sqrt{3}}{2}$$

$$\boxed{y=C_1+C_2x+C_3x^2+C_4e^{-2x}+e^{-\frac{3}{2}x}\left(C_5\cos\left(\frac{\sqrt{3}}{2}x\right)+C_6\sin\left(\frac{\sqrt{3}}{2}x\right)\right)}$$

Choose only one from the following

b)



The differential equation for an RLC circuit is presented as:

$$L\frac{d^2i}{dt^2}+R\frac{di}{dt}+\frac{1}{C}i=0$$

Suppose an RLC circuit with a resistor $R=10\ \Omega$, an inductor $L=0.5\ H$ and Capacitor $C=0.02\ F$ is connected to a DC voltage source. Find the current $i\left(t\right)$. **(4 points)**

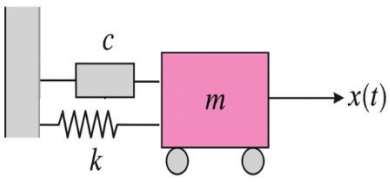
$$0.5\frac{d^2i}{dt^2}+10\frac{di}{dt}+50i=0$$

$$0.5m^2+10m+50=0$$

$$m_1=m_2=-10$$

$$\boxed{i=C_1e^{-10t}+C_2te^{-10t}}$$

b)



The mathematical model for the mass–spring–damper system for free vibration is presented as the following ODE:

$$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=0$$

Given $m=0.5\ Kg$, $c=10\ \ N.s/m$, $k=100\ N/m$. Find the the displacement $x(t)$ of the mass from its equilibrium position **(4 points)**

$$0.5\frac{d^2x}{dt^2}+10\frac{dx}{dt}+100x=0$$

$$0.5m^2+10m+100=0$$

$$m_{1,2}=-10\pm10i$$

$$\boxed{x=e^{-10t}\left(C_1\cos\left(10t\right)+C_2\sin\left(10t\right)\right)}$$

GOOD LUCK